

Progress Test 2.C — Thermo-S1

2.C · Thermoelastic coefficients: principle and method

Semester 1 · Academic Year 2026–2027

ANSWER KEY · MODEL ANSWERS AND MARKING SCHEME

Duration: 10 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Quick calculation

(1 mark)

At $p = 1.0 \times 10^5 \text{ Pa}$, a gas has $\alpha = 3.3 \times 10^{-3} \text{ K}^{-1}$ and $\chi_T = 1.0 \times 10^{-5} \text{ Pa}^{-1}$. Give its piezothermic coefficient β .

$$\beta = \frac{\alpha}{p\chi_T} = \frac{3.3 \times 10^{-3}}{1.0 \times 10^5 \times 1.0 \times 10^{-5}} \quad \boxed{\beta = 3.3 \times 10^{-3} \text{ K}^{-1}}$$

Marking

①
0 1

Marking scheme

① 1 pt $\beta = 3.3 \times 10^{-3} \text{ K}^{-1}$ (the product $p\chi_T$ is dimensionless).

Exercise 2. Relation between the coefficients

(5 marks)

State the cyclic relation between the partial derivatives of p , V and T . Use it to show that $\beta = \frac{\alpha}{p\chi_T}$.

For the equation of state $f(p, V, T) = 0$:

$$\left(\frac{\partial p}{\partial T}\right)_V \left(\frac{\partial T}{\partial V}\right)_p \left(\frac{\partial V}{\partial p}\right)_T = -1$$

With $\left(\frac{\partial T}{\partial V}\right)_p = 1/\left(\frac{\partial V}{\partial T}\right)_p$:

$$\left(\frac{\partial p}{\partial T}\right)_V = -\frac{(\partial V/\partial T)_p}{(\partial V/\partial p)_T} = -\frac{\alpha V}{-\chi_T V} = \frac{\alpha}{\chi_T}$$

Hence $\beta = \frac{1}{p} \left(\frac{\partial p}{\partial T}\right)_V$ gives $\boxed{\beta = \frac{\alpha}{p\chi_T}}$

Marking

①
0 1 2
②
0 1 2
③
0 1

Marking scheme

- ① 2 pts Cyclic relation, with the correct fixed variables and -1 on the right.
 ② 2 pts $(\partial p/\partial T)_V = \alpha/\chi_T$, derived.
 ③ 1 pt $\beta = \alpha/(p\chi_T)$.

Examiner's tip

The product of the three derivatives is -1 , not $+1$. Check the sign on an ideal gas: $\alpha = 1/T$ and $\chi_T = 1/p$ give $\beta = 1/T > 0$.

Exercise 3. Method in thermodynamics

(3 marks)

To obtain the equation of state from α and χ_T , one must:

- ☐ differentiate ☒ integrate ☐ multiply them ☐ average them

Why: knowing α and χ_T as functions of (T, p) , a double integration rebuilds $V(T, p)$ (Fig. 8 of the notes).

To obtain α from a known equation of state, one must:

- ☒ differentiate ☐ integrate ☐ invert it ☐ plot it

Why: express $V(T, p)$, then compute the partial derivative $(\partial V / \partial T)_p$.

The three coefficients α , χ_T and β are:

- ☐ independent of each other ☒ linked by $\beta = \alpha / (p\chi_T)$
☐ always equal ☐ all dimensionless

Why: only two of them are independent; the third follows from the cyclic relation.

Exercise 4. From an equation of state to the coefficients

(5 marks)

A gas obeys the equation of state $p(V - nb) = nRT$, where the co-volume b is a positive constant. Compute α , χ_T and β , then check that $\beta = \alpha / (p\chi_T)$. What do you find when $b \rightarrow 0$?

$V = \frac{nRT}{p} + nb:$	$\left(\frac{\partial V}{\partial T}\right)_p = \frac{nR}{p}$ so	$\alpha = \frac{nR}{pV}$	$\left(\frac{\partial V}{\partial p}\right)_T = -\frac{nRT}{p^2}$ so
$\chi_T = \frac{nRT}{p^2V}$			
$p = \frac{nRT}{V - nb}:$	$\left(\frac{\partial p}{\partial T}\right)_V = \frac{nR}{V - nb} = \frac{p}{T}$ so	$\beta = \frac{1}{T}$	
Check: $\frac{\alpha}{p\chi_T} = \frac{nR}{pV} \times \frac{p^2V}{pnRT} = \frac{1}{T} = \beta. \checkmark$			
When $b \rightarrow 0$: $V = nRT/p$, so $\alpha = \frac{1}{T}$ and $\chi_T = \frac{1}{p}$, the values for the ideal gas (3.A).			

Marking

①
0 1
②
0 1
③
0 1
④
0 1
⑤
0 1

Marking scheme

- ① 1 pt $\alpha = nR/(pV)$.
 ② 1 pt $\chi_T = nRT/(p^2V)$.
 ③ 1 pt $\beta = 1/T$.
 ④ 1 pt Relation checked.
 ⑤ 1 pt Limit $b \rightarrow 0$: $\alpha = 1/T$, $\chi_T = 1/p$.

Your score: / 14 $\rightarrow$ mark out of 20 = score $\times$ 20/14 = / 20